4.4.2. Weekly Graded Assessment

Spatial Filtering

3/3 points (graded)
Which of the following statements are NOT true?

- [x] Fiji/ImageJ can handle odd and even filter kernels.
- [ ] With an odd filter kernel no interpolation is necessary.
- [ ] Normalizing filter kernels is a way to prevent global intensity changes in the image after filtering.
- [x] Fine details within an image are enhanced (better visible) after filtering with a classical mean filter.
- [x] Spatial filters are not suited to enhance the edges of an object.

✔

Submit You have used 1 of 2 attempts

✔ Correct (3/3 points)

Image Filtering

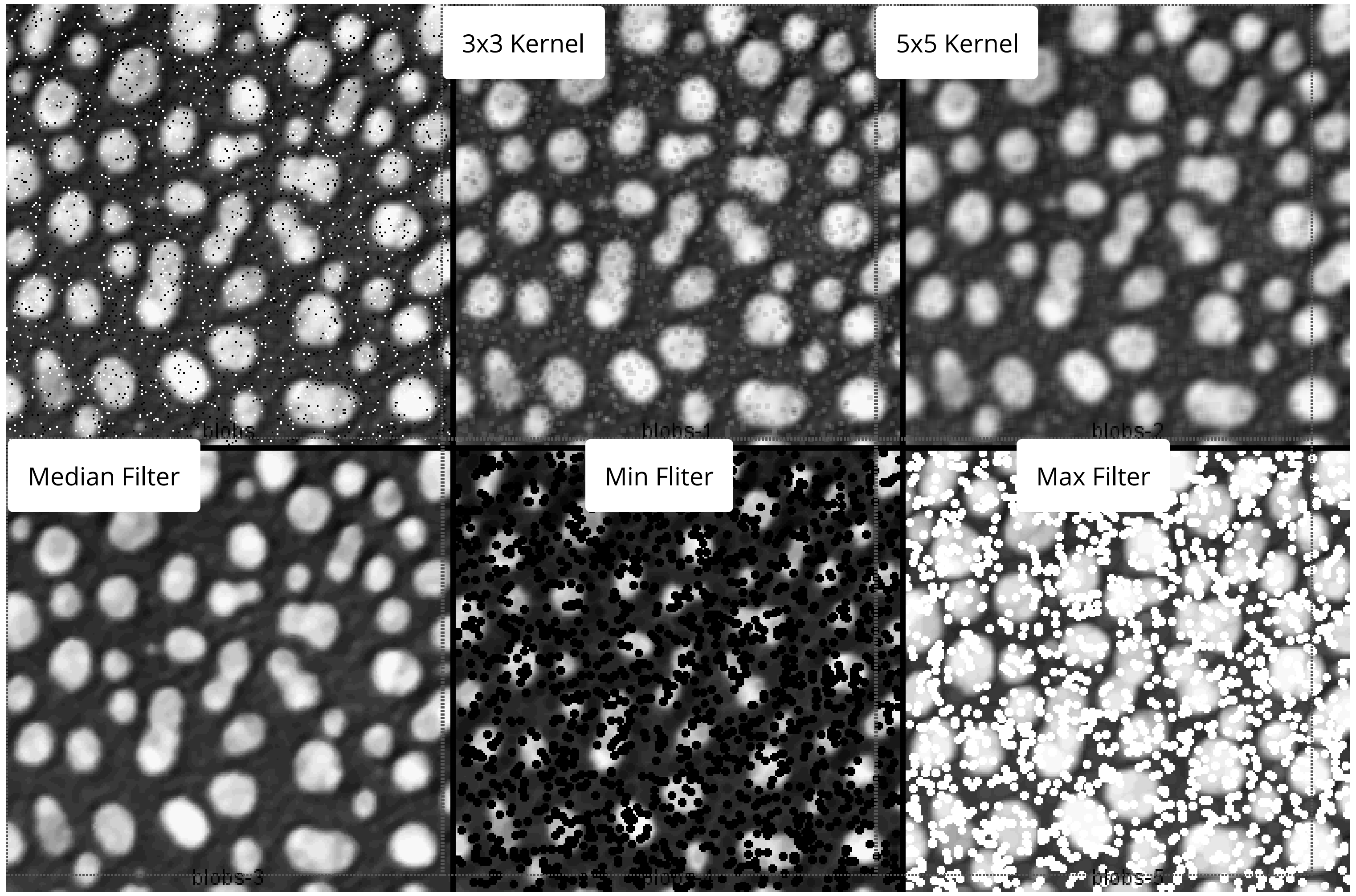
4/4 points (graded)

Keyboard Help

PROBLEM

Drag the items onto the image above. The original is shown in the upper-left corner. The other images are the results of a filter operation.
To obtain the image used here, you can open **Blobs**, invert the Lookup Table and apply Salt & Pepper

noise (**Process > Noise > Salt And Pepper**)



Submit You have used 1 of 2 attempts.

Reset Show Answer

FEEDBACK

✔ Correctly placed 5 items.

✔ Good work! You have completed this drag and drop problem.

✔ Your highest score is 4.0

Fourier Filtering

3/3 points (graded)

Which of the following statements are true?

- [x] Fourier filtering can help enhance or reduce periodic (repeating) signals in images.
- [x] You can move from the Fourier domain to the spatial domain by applying the Inverse Fourier Transform.
- [x] The Fourier transform of an image is always square and the width/height is always a power of 2 (you can test this in Fiji).
- [ ] The closer we are from the center of the Fourier image, the higher the frequency.

Submit You have used 2 of 2 attempts

Correct (3/3 points)

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